

## Solutions to JEE Advanced – 4 | JEE 2023 | Paper-1 (Gen 1 to 6)

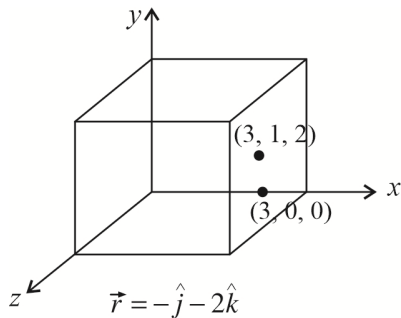
## PHYSICS

- 1.(A) Say final temperature
- $T$
- is.

$$\begin{aligned}
 1 \times 2 \times 20 + \frac{1}{2} \times (6+2) \times 20 + (T-40)6 \\
 = 2 \times 20 + \frac{1}{2} \times (6+2) \times 10 + (70-T)6 \\
 80 + 6T - 240 = 40 + 420 - 6T \\
 12T = 420 + 240 - 40 \\
 T = 620/12; \quad T = 51.6^\circ\text{C}
 \end{aligned}$$

$$2.(A) \quad D_{\min} = \sqrt{5} \quad |v| = \frac{\sqrt{5}}{2}$$

$$\vec{v} = \frac{\sqrt{5}}{2} \left( \frac{-\hat{j} - 2\hat{k}}{\sqrt{5}} \right) = \frac{-\hat{j} - 2\hat{k}}{2}$$



- 3.(C) String becomes taut again

$$4t = 2t + \frac{1}{2}gt^2$$

$$2t = 5t^2 \quad \text{or} \quad t = 0.4$$

$$(3) \quad 3v_c = 2 \times 4 + 1 \times (2 + 10 \times 0.4) \text{ by momentum conservation}$$

$$3v_c = 8 + 6 = 14$$

$$v_c = \frac{14}{3} \text{ m/s}$$

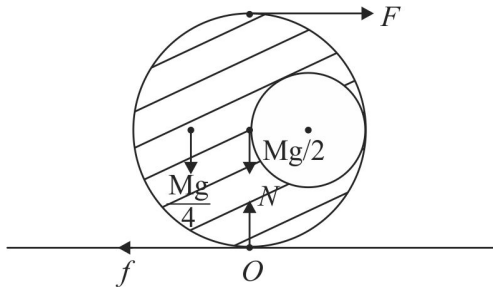
- 4.(C) For minimum
- $v_0$
- , particle just graze the surface of earth. Orbit will be elliptical.

$$mv_0 3R = mvR \Rightarrow v = 3v_0$$

$$\frac{1}{2}mv_0^2 - \frac{GMm}{3R} = \frac{1}{2}m(9v_0^2) - \frac{GMm}{R}$$

$$\frac{GMm}{R} \left( \frac{2}{3} \right) = \frac{1}{2}mv_0^2 8; \quad v_0 = \sqrt{\frac{GM}{6R}}$$

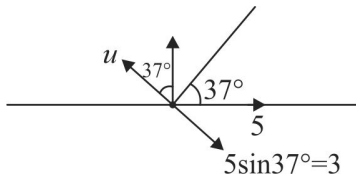
5.(D)



$$\tau_0 = 0 \Rightarrow \frac{Mg}{4} \times \frac{R}{2} = F(2R)$$

$$F = \frac{Mg}{16}$$

6.(D)



$$t = \frac{500}{3 \cos 37^\circ} + \frac{500}{3 \cos 37^\circ}$$

$$= \frac{1000}{3 \times \frac{4}{5}} = \frac{250 \times 5}{3} = \frac{1250}{3} \text{ s}$$

7.(CD)  $\Rightarrow v' \cos \theta = v; \quad v' = \frac{v}{\cos \theta} = \frac{5v}{4}$

$$\Rightarrow a = a' \cos \theta + v'(-\sin \theta) \frac{d\theta}{dt}$$

$$a = a' \cos \theta - v' \sin \theta \frac{d\theta}{dt} \quad \left( \because \cos \theta = \frac{4}{5} \right)$$

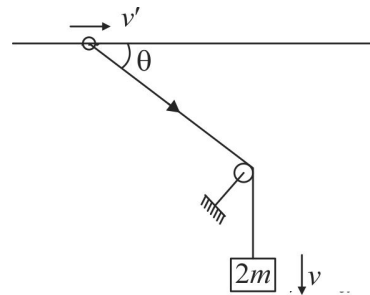
$$\Rightarrow a < \frac{4a'}{5}$$

$$\Rightarrow N = mg + T \sin \theta$$

$$2mg - T = 2ma$$

$$N = mg + (2mg - 2ma) \frac{3}{5}$$

$$= \frac{5mg + 6mg - 6ma}{5} = \frac{11mg - 6ma}{5}$$



8.(ACD)  $\mu mg = ma$

$$F_0 - 4\mu mg = 4ma$$

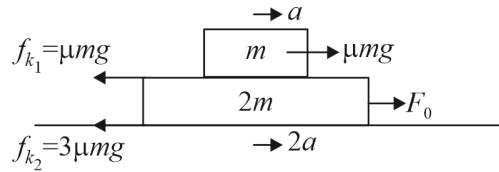
$$F_0 = 8ma \text{ or } a = \frac{F_0}{8m}$$

$$\mu = \frac{F_0}{8mg}$$

$$W_{f_{k_2}} = -\mu(3mg) \times \frac{1}{2} \frac{F_0}{4m} t^2 = -\frac{3\mu g F_0 t^2}{8}$$

$$W_{f_{k_1}} = -f_{k_1} \frac{1}{2} (2a)t^2 + f_{k_1} \frac{1}{2} at^2 = -f_{k_1} \times \frac{1}{2} at^2 = -\mu mg \times \frac{1}{2} \frac{F_0}{8m} t^2$$

$$P_{av} = \frac{W}{t} = -\frac{F_0 t^2 \mu g}{16t} = -\frac{F_0 \mu g}{16} t$$



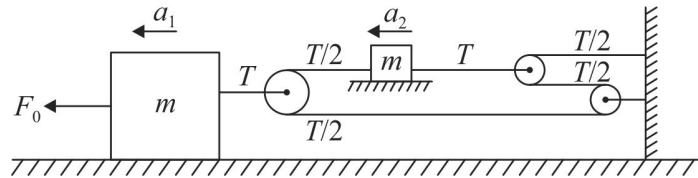
9.(BD)  $\Rightarrow -Ta_1 - \frac{T}{2}a_2 = 0$

$$\Rightarrow a_2 + 2a_1 = 0$$

$$\Rightarrow F_0 - T = ma_1$$

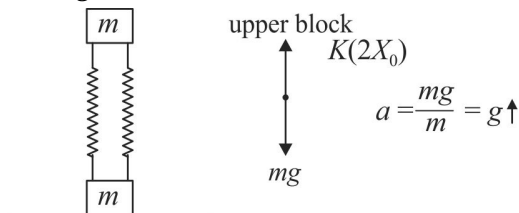
$$\frac{T}{2} = -ma_2 = 2ma_1$$

by solving  $a_1 = \frac{F_0}{5m}; T = \frac{4F_0}{5}$

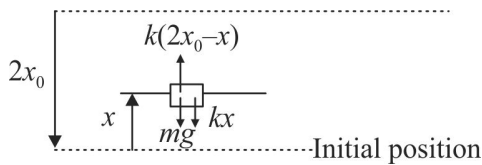


10.(BCD)

Just after string is cut



Upper block will have max speed when  $a = 0$



$$k(2x_0 - x) = mg + kx$$

$$2mg - kx = mg + kx$$

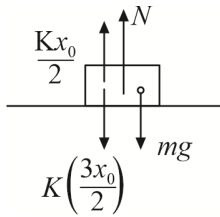
$$x = \frac{mg}{2k} = \frac{x_0}{2} \quad (\because mg = kx_0)$$

By conservation of energy :

$$mgx + \frac{1}{2} kx^2 + \frac{1}{2} mv^2 = -\frac{1}{2} k(2x_0 - x)^2 + \frac{1}{2} k(2x_0)^2$$

$$v = x_0 \sqrt{\frac{k}{2m}}$$

Lower block



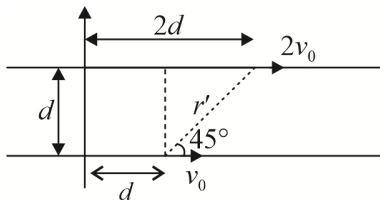
$$\frac{Kx_0}{2} + N = \frac{3Kx_0}{2} + mg; \quad N = mg + mg = 2mg$$

At max elongation (K.E. = 0)

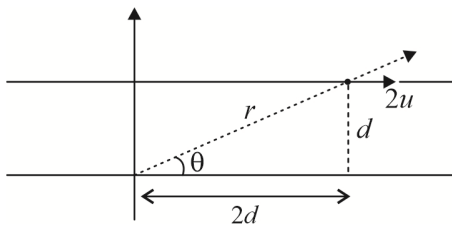
$$mgy + \frac{1}{2}ky^2 = \frac{1}{2}k(2x_0)^2 - \frac{1}{2}k(2x_0 - y)^2; \quad mgy + \frac{ky^2}{2} = \frac{k}{2}(4x_0y - y^2)$$

$$mg = \frac{k}{2}(4x_0 - 2y); \quad mg = 2Kx_0 - ky \Rightarrow \quad ky = mg; \quad y = \frac{mg}{k}$$

11.(ABC)



$$\Rightarrow \quad \omega = \frac{V_0 \cos 45^\circ}{\sqrt{2}d} = \frac{v_0}{2d} \Rightarrow \quad \frac{dr'}{dt} = \frac{2v_0}{\sqrt{2}} - \frac{v_0}{\sqrt{2}} = \frac{v_0}{\sqrt{2}}$$



$$\tan \theta = \frac{1}{2}$$

$$\frac{dr}{dt} = 2v_0 \cos \theta = 2v_0 \cdot \frac{2}{\sqrt{5}} = \frac{4v_0}{\sqrt{5}}$$

12.(ABCD)

$$\Rightarrow \quad Q = a\sqrt{2gh_1}; \quad h_1 = \frac{Q^2}{a^2(2g)} = \frac{H}{2}$$

$$\Rightarrow \quad F = \rho \left( \sqrt{2g \frac{H}{4}} \right)^2 a = \frac{\rho g H a}{2} \quad (\because F = \rho A V^2)$$

$$\Rightarrow \quad -A \frac{dh}{dt} = a\sqrt{2gh}; \quad \int_0^t dt = \frac{-A}{a\sqrt{2g}} \int_{H/4}^0 \frac{dh}{\sqrt{h}}$$

$$t = \frac{A}{a\sqrt{2g}} \sqrt{\frac{H}{4}} = \frac{A\sqrt{H}}{2a\sqrt{2g}}$$

$$\Rightarrow \quad \text{If tap is open level above hole will be } \frac{H}{2} \text{ so level from bottom} = \frac{3H}{4}$$

1.(0.39)

$$2.(6) \Rightarrow F = -\frac{GM}{2R^2} + \frac{G(2M)}{\left(\frac{3R}{2}\right)^2} = 1 \times a \quad \left( \vec{g} = -\frac{4\pi G\delta}{3} \overrightarrow{OO'} = \frac{GM}{2R^2} (-\hat{i}) \right)$$

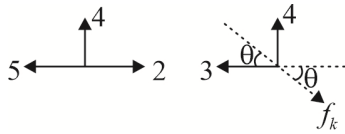
$$a = -\frac{GM}{2R^2} + \frac{8GM}{9R^2} = \frac{(-9+16)GM}{18R^2} = \frac{7GM}{18R^2} = 0.39 \frac{GM}{R^2}$$

$$\Rightarrow \frac{1}{2}mv^2 = -\frac{mGM}{2R^2} \times \frac{R}{2} + \left( \frac{2G(M/2)}{3R} - \frac{2GM}{2R} \right)$$

$$\frac{v^2}{2} = \frac{GM}{R} \left( -\frac{1}{4} + \frac{1}{3} \right) = \frac{GM}{12R}; \quad v = \sqrt{\frac{GM}{6R}}$$

3.(53), 4.(12.50)

$v_{\text{block/mat}}$  :



$$\tan \theta = \frac{4}{3}; \quad \theta = 53^\circ$$

$$W_{fk} = \Delta k \quad (\text{w.r.t. mat})$$

$$-\mu mgd = \frac{1}{2}m(0)^2 - \frac{1}{2}m(5)^2; \quad d = \frac{25}{2 \times \mu g} = 12.5\text{m}$$

5.(10), 6.(9.65)

$$N\Delta t = \mu u(1+e)$$

$$\mu(N\Delta t) = \mu u_x \quad \left( \because \frac{\mu N\Delta t}{m} = u_x = \frac{1}{3} \times u \times \frac{3}{2} = \frac{u}{2} \right)$$

$$\mu \mu u(1+e) = \mu u_x$$

$$u_x = \mu(u)(1+e).$$

$$\text{Range} = \frac{2u_y u_x}{g} = \frac{2eu}{g} \mu u(1+e) = \frac{2\mu e u^2(1+e)}{g}$$

$$= \frac{2 \times \frac{1}{3} \times \frac{1}{2} \times u^2 \times \frac{3}{2}}{g} = \frac{u^2}{2g} = \frac{2gh}{2g} = h = 10\text{m}$$

$$-(\mu N\Delta t)R = I(\omega_2 - \omega_0)$$

$$\omega_2 = \omega_0 - \frac{\mu NR\Delta t}{\frac{2}{5}mR^2} = \omega_0 - \frac{5}{2} \times \frac{1}{3} \times \frac{3}{2} \frac{u}{R} = \omega_0 - \frac{5u}{4R} = 45 - 25\sqrt{2} = 9.65 \frac{\text{rad}}{\text{s}}$$

## CHEMISTRY

- 1.(A)  $\text{HCl} \Rightarrow 5 \text{ m moles}$   
 $\text{NaOH} \Rightarrow 30 \text{ m moles}$   
 Remaining  $\text{NaOH} \Rightarrow 25 \text{ m moles}$   
 $2\text{NaOH} + \text{CO}_2 \longrightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$   
 $25 \text{ m moles} \quad 12.5 \text{ m moles}$   
 $V_{\text{CO}_2} = 12.5 \times 10^{-3} [22400] = 12.5 \times 22.4 = 280 \text{ mL}$
- 2.(B) Total number of spectral lines  

$$N = \frac{n(n-1)}{2}$$

$$10 = \frac{n(n-1)}{2}$$

$$20 = n(n-1) \Rightarrow n = 5$$
 de-Broglie wavelength  

$$\lambda = 2\pi a_0 \left( \frac{n}{Z} \right)$$

$$\lambda = 10\pi a_0 \quad (n = 5, Z = 1)$$
 $(a_0 = \text{Bohr's radii})$
- 3.(C) Correct order  
 (A)  $\text{Ga} < \text{Al} < \text{In}$       (B)  $\text{Ti} < \text{Zr} \approx \text{Hf}$       (C)  $\text{C} > \text{N} > \text{O}$   
 Option (C) is correct
- 4.(D)  $[\text{HCl}] = 10^{-8} \text{ M}$ ; pH is just less than 7; pOH is just greater than 7  
 $[\text{NaOH}] = 10^{-8} \text{ M}$ ; pOH is just less than 7  
 $[\text{NaHCO}_3] = 10^{-2} \text{ M}$ ;  $\text{pH} = \frac{6+9}{2} = 7.5$ ,  $\text{pOH} = 6.5$ , for equimolar mixture of  $\text{CH}_3\text{COOH}$  and  $\text{CH}_3\text{COONa}$ ;  $\text{pH} = 4.7$  thus  $\text{pOH} = 9.3$  at  $25^\circ\text{C}$ .
- 5.(A) V.S. of  $\text{H}_2\text{O}_2 = 11.2 \times \text{molarity}$   
 $5.6 = 11.2 \times \text{molarity}$   
 $\therefore \text{Molarity of } \text{H}_2\text{O}_2 = 0.5 \text{ M}$   
 By law of chemical equivalence  $10 \times 0.5 \times 2 = 20 \times M \times 5$   
 $\Rightarrow M = \text{Molarity of } \text{KMnO}_4 = 0.1 \text{ M}$
- 6.(C) 
$$\text{I}_2 \xrightarrow{\text{OH}^-} \text{I}^- + \text{X}$$

|               |     |        |       |
|---------------|-----|--------|-------|
| Given         | (2) | (10/3) | (2/3) |
| $\Rightarrow$ | (6) | (10)   | (2)   |
| or            | 3   | 5      | 1     |

 $\Rightarrow 3 \text{ moles of } \text{I}_2 \text{ undergo disproportionation into 5 moles of } \text{I}^- \text{. Thus } \text{X} = \text{IO}_3^- (+5 \text{ state of iodine})$

- 7.(AC)  $\text{CO} \rightarrow$  Diamagnetic  
 $\text{NO} \rightarrow$  Paramagnetic  
 $\text{C}_2 \rightarrow$  Diamagnetic  
 $\text{B}_2 \rightarrow$  Paramagnetic

8.(ABC)

$$\Delta_r H^\circ = \Delta_f H^\circ_{(P)} - \Delta_f H^\circ_{(A)} = -80 - [-100] = +20 \text{ kJ}$$

$$\Delta_r S^\circ = S^\circ_P - S^\circ_A = 20 - 10 = 10 \text{ J / kelvin-mole, now at 300 K}$$

$$\Delta_r G^\circ = \Delta_r H^\circ - T\Delta_r S^\circ = 20 - \frac{10 \times 300}{1000} = +17 \text{ kJ}$$

The endothermic reaction will be spontaneous above its equilibrium temperature.

$$T_{\text{eq}} = \frac{\Delta_r H^\circ}{\Delta_r S^\circ} = \frac{20 \times 1000}{10} = 2000 \text{ Kelvin}$$

9.(ACD)

- (A)  $|W_{\text{sys}}|_{\text{rev}} > |W_{\text{sys}}|_{\text{irrev}}$   
 (B)  $q$  : path function,  $\Delta H$  : State function  
 (C)  $|W_{\text{Isobaric}}| > |W_{\text{Isothermal}}|$   
 (D)  $\Delta S_{\text{sys}}$  is NOT always positive for spontaneous process

10.(ABC)

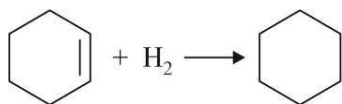
(A), (B), (C) constitute pairs of constitutional/structural isomers.  
 The compounds in option (D) are NOT isomers.

11.(BD) Statements of option B and D are correct; according to K.T.G.

12.(ACD)

Statements of option A, C and D are NOT true according to the Le Chatelier's principle.

1.(200)



$$\Delta_r H = \Delta_f H^\circ_{(\text{cyclohexane})} - \Delta_f H^\circ_{(\text{cyclohexene})} = 80 - [x] = -120; x = 200 \text{ kJ / mole}$$

2.(-152)

$$\begin{aligned} \text{R.E of benzene} &= 3[\Delta H^\circ_{\text{Hydg.}}]_{(\text{cyclohexene})} - [\Delta H^\circ_{\text{Hydg.}}]_{(\text{benzene})} \\ &= 3[-120] - [-208] = -360 + 208 = -152 \text{ kJ / mole} \end{aligned}$$

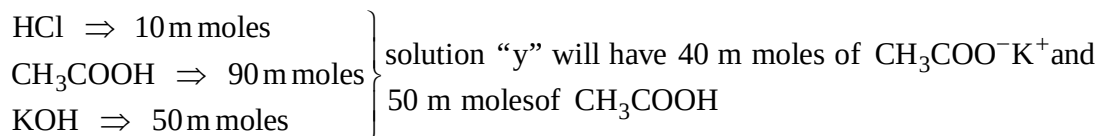
3.(2.35)

$$6\% \text{ w/v } \text{CH}_3\text{COOH} \Rightarrow \text{Molarity} = \frac{10 \times 6}{60} = 1\text{M}$$

$$K_a = \frac{[\text{H}^+]^2}{[\text{CH}_3\text{COOH}]} = \frac{x^2}{1} = x^2 = 2 \times 10^{-5}; \text{pH} = \frac{1}{2}[\text{p}K_a] = 2.35$$

4.(4.60)

Final mixture



$$\text{pH} = \text{p}K_a + \log_{10} \frac{[\text{CH}_3\text{COO}^-\text{K}^+]}{[\text{CH}_3\text{COOH}]} = 4.7 + \log_{10} \frac{4}{5} = 4.7 + [0.60 - 0.70] = 4.60$$

5.(4) The given compound have 4 F.G.

$$\rightarrow \text{amide} \quad \rightarrow \text{anhydride} \quad \rightarrow \text{acid} \quad \rightarrow \text{ester}$$
6.(15) D.O.U = number of  $\pi$ -bonds + number of rings =  $8 + 7 = 15$ 

## MATHEMATICS



1.(B)  $|16z_1z_2z_3 + 9z_1z_2z_4 + 4z_1z_3z_4 + z_2z_3z_4| = 48$

$$|z_1z_2z_3z_4| \left| \frac{16}{z_4} + \frac{9}{z_3} + \frac{4}{z_2} + \frac{1}{z_1} \right| = 48$$

$$\left| \frac{1}{z_1} + \frac{4}{z_2} + \frac{9}{z_3} + \frac{16}{z_4} \right| = \frac{48}{|z_1||z_2||z_3||z_4|}$$

$$\left( \frac{1}{z_1} + \frac{4}{z_2} + \frac{9}{z_3} + \frac{16}{z_4} \right) = 2$$

2.(C)  $A = \left( (5 + 2\sqrt{6})^2 \right)^{1/4} = (5 + 2\sqrt{6})^{1/2} = \left[ (\sqrt{3} + \sqrt{2})^2 \right]^{1/2}$

Hence,  $A = \sqrt{3} + \sqrt{2}$ ,  $B = 8\sqrt{3} + \frac{8\sqrt{6}}{\sqrt{3}} + \frac{16}{\sqrt{3}} + \dots \infty$

$$r = \frac{8\sqrt{6}}{\sqrt{3}} \times \frac{1}{8\sqrt{3}} = \sqrt{\frac{2}{3}}, \quad \therefore B = \frac{8\sqrt{3}}{1 - (\sqrt{2}/\sqrt{3})} = \frac{(8\sqrt{3})\sqrt{3}}{\sqrt{3} - \sqrt{2}}$$

$$B = 24(\sqrt{3} + \sqrt{2})$$

$$(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2})x^2 + \frac{24(\sqrt{3} + \sqrt{2})}{(\sqrt{3} + \sqrt{2})}x + C = 0$$

$$x^2 + 24x + C = 0 \quad \dots(i)$$

Now,  $|\alpha - \beta| = (6\sqrt{6})^k$

$$k = \log_6 10 - \log_6 5 + \frac{1}{2} \log_6 (\log_6 (18 \times 72))$$

$$= \log_6 2 + \frac{1}{2} \log_6 (\log_6 1296) = \log_6 2 + \frac{1}{2} \log_6 4 = 2 \log_6 2 = \log_6 4$$

$$\therefore |\alpha - \beta| = (6\sqrt{6})^{\log_6 4} = \left( (6)^{3/2} \right)^{2 \log_6 2} = 6^{\log_6 8} = 8$$

Hence,  $(\alpha - \beta)^2 = 64$

$$(\alpha + \beta)^2 - 4\alpha\beta = 64$$

$$576 - 4C = 64 \Rightarrow 4C = 512 \Rightarrow C = 128$$

3.(D)  $x_1^2 + \sqrt{x_2^2 - 2x_2} = 2x_1 - 1 \quad \dots(i)$

$$\Rightarrow (x_1 - 1)^2 = -\sqrt{x_2(x_2 - 2)} \quad \dots(ii)$$

Notice that  $(x_1 - 1)^2 \geq 0$  and  $-\sqrt{x_2(x_2 - 2)} \leq 0$

So, this will force to deduce that  $(x_1 - 1)^2 = 0$  and  $-\sqrt{x_2(x_2 - 2)} = 0$

For equation (i)  $(x_1 - 1)^2 = 0 \Rightarrow x_1 = 1$

For equation (ii)  $-\sqrt{x_2(x_2 - 2)} = 0 \Rightarrow x_2 = 0$  or  $x_2 = 2$

But  $x_2$  is positive real solution  $\Rightarrow x_2 = 2$

Thus,  $x^2 - bx + c = (x-1)(x-2) = 0 \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow b = 3$  and  $c = 2$

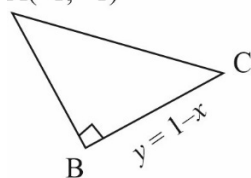
$$\Rightarrow b + c = 3 + 2 = 5 \Rightarrow (b + c)_{\min} = 5$$

4.(B)  $x^3 + y^3 + (-1)^3 - 3(-1)(x)(y) = 0$

$$(x + y - 1) \left[ (x - y)^2 + (y + 1)^2 + (x + 1)^2 \right] = 0$$

Hence,  $A \equiv (-1, -1)$  and the equation of side BC is  $x + y = 1$

$A(-1, -1)$



$(a, 1-a)$

$$m_{AB} \cdot m_{BC} = -1 \Rightarrow \left( \frac{2-a}{a+1} \right) (-1) = -1$$

$$\therefore 2 - a = a + 1 \Rightarrow a = \frac{1}{2}; B = \left( \frac{1}{2}, \frac{1}{2} \right)$$

Hence, equation of AB is  $y = x$

$m = 1$  and  $c = 0$

Hence,  $4 - m - c = 4 - 1 - 0 = 3$

5.(B) Let  $\frac{a+b}{c} = t$

$$\therefore (a+b)^2 = (a+c)(b+c) \text{ and } c = \frac{2ab}{a+b}$$

$$= ab + (a+b)c + c^2 \Rightarrow (a+b)^2 = \frac{(a+b)c}{2} + (a+b)c + c^2$$

Dividing by  $c^2$ , we get  $\left( \frac{a+b}{c} \right)^2 = \frac{3}{2} \left( \frac{a+b}{c} \right) + 1$

$$\therefore 2t^2 - 3t - 2 = 0 \Rightarrow (t-2)(2t+1) = 0 \Rightarrow t = 2$$

6.(A) Using  $x + \frac{1}{y} \geq 2\sqrt{\frac{x}{y}}$  we get

$$4 = x_1 + \frac{1}{x_2} \geq 2\sqrt{\frac{x_1}{x_2}}$$

$$1 = x_2 + \frac{1}{x_3} \geq 2\sqrt{\frac{x_2}{x_3}}; 4 = x_3 + \frac{1}{x_4} \geq 2\sqrt{\frac{x_3}{x_4}}$$

$$1 = x_4 + \frac{1}{x_1} \geq 2\sqrt{\frac{x_4}{x_1}}; 16 = \left( x_1 + \frac{1}{x_2} \right) \left( x_2 + \frac{1}{x_3} \right) \left( x_3 + \frac{1}{x_4} \right) \left( x_4 + \frac{1}{x_1} \right) \geq 2^4$$

This is only possible when

$$x_1 = \frac{1}{x_2}, x_2 = \frac{1}{x_3}, x_3 = \frac{1}{x_4} \text{ and } x_4 = \frac{1}{x_1}$$

$$\therefore x_1 = 2, x_2 = \frac{1}{2}, x_3 = 2 \text{ and } x_4 = \frac{1}{2}$$

7.(AB) Let  $a, b$ , and  $c$  denote the number of  $-1s, 1s$ , and  $2s$  in the sequence, respectively. We need not consider the zeros. Then  $a, b, c$  are nonnegative integers satisfying  $-a + b + 2c = 19$  and  $a + b + 4c = 99$ .

It follows that  $a = 40 - c$  and  $b = 59 - 3c$ , where  $b = 59 - 3c, 0 \leq c \leq 19$  (since  $b \geq 0$ ),

$$\text{so } x_1^3 + x_2^3 + \dots + x_n^3 = -a + b + 8c = 19 + 6c.$$

When  $c = 0$  ( $a = 40, b = 59$ ), the lower bound (19) is achieved.

When  $c = 19$  ( $a = 21, b = 2$ ), the upper bound (133) is achieved.

8.(AC) Let person  $P_i$  get  $x_i$  number of things such that  $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 30$

If  $x_i$  is odd or  $x_i = 2\lambda_i + 1$ , where  $\lambda_i \geq 0$  then  $2(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6) + 6 = 30$

$$\text{or } \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6 = 12$$

Then number of solutions  $^{12+6-1}C_{6-1} = ^{17}C_5$

If  $x_i$  is even or  $x_i = 2\lambda_i$ , where  $\lambda_i \geq 1$ , then :  $2(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6) = 30$

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6 = 15$$

Therefore, required number of ways is  $^{15-1}C_{6-1} = ^{14}C_5$

9.(ABCD)

Assume  $a = b$ , we have nothing to prove.

$$(A) \quad \text{for } a > b; \quad \sqrt{a} - \sqrt{b} = \frac{a-b}{\sqrt{a} + \sqrt{b}} \leq \frac{a-b}{\sqrt{a}} < \frac{a-b}{\sqrt{a-b}} \quad \text{Hence option (A) is correct}$$

$$(B) \quad \text{The given inequality is equivalent to } a^3 + b^3 \geq ab(a+b)$$

$$\text{For } a < b \text{ or } a > b \Rightarrow a^2 + b^2 - ab \geq ab \Rightarrow (a-b)^2 \geq 0 \text{ for all } a > 0, b > 0$$

Hence option (B) is also correct.

(C) The given inequality is equivalent to

$$a^{3/2} + b^{3/2} \geq \sqrt{ab}(\sqrt{a} + \sqrt{b})$$

$$\text{For } a < b \text{ or } a > b \Rightarrow (a^{1/2})^2 + (b^{1/2})^2 - a^{1/2}b^{1/2} \geq \sqrt{ab} \Rightarrow (\sqrt{a} - \sqrt{b})^2 \geq 0$$

Hence option (C) is also correct.

(D) Given expression is equivalent to

$$\frac{2a^2}{1+a^4} + \frac{2b^2}{1+b^4} \leq 2 \Rightarrow \frac{2a^2}{1+a^4} - 1 + \frac{2b^2}{1+b^4} - 1 \leq 0 \Rightarrow \frac{(1-a^2)^2}{1+a^4} + \frac{(1-b^2)^2}{1+b^4} \geq 0$$

$$\text{As } (1-a^2) \geq 0 \text{ for all } a > 0 \quad \text{So } \frac{(1-a^2)^2}{1+a^4} \geq 0 \text{ for all } a > 0$$

Similarly  $\frac{(1-b^2)^2}{1+b^4} \geq 0$  for all  $b > 0$  Hence option (D) is also correct.

10.(ABD)

We have  $\sin x \sqrt{8 \cos^2 x} = 1$

$$\Rightarrow \sin x |\cos x| = \frac{1}{2\sqrt{2}}$$

Case-I : When  $\cos x > 0$

$$\text{In this case } \sin x \cos x = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow \sin 2x = +\frac{1}{\sqrt{2}} \Rightarrow 2x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4} \Rightarrow x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}$$

As  $x$  lies between  $0$  and  $2\pi$  and  $\cos x > 0$

$$\therefore x = \frac{\pi}{8}, \frac{3\pi}{8}$$

Case-II : When  $\cos x < 0$ .

$$\text{In the case } \sin x |\cos x| = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow \sin x \cos x = -\frac{1}{2\sqrt{2}} \text{ of } \sin 2x = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow x = \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8} \Rightarrow x = \frac{5\pi}{8}, \frac{7\pi}{8} \text{ as } \cos x < 0$$

Thus the values of  $x$  satisfying the given equation which lie between  $0$  and  $2\pi$  are  $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$

These are in A.P. with common difference  $\frac{\pi}{4}$

$$11.(ABD) \quad (1+x^2+x^4)^8 = C_0 + C_1x^2 + C_2x^4 + \dots + C_{16}x^{32} \quad \dots(i)$$

Putting  $z = i$ , where  $i = \sqrt{-1}$ .

$$(1-1+1)^8 = C_0 - C_1 + C_2 - C_3 + \dots + C_{16} \Rightarrow C_0 - C_1 + C_2 - C_3 + \dots + C_{16} = 1$$

Also, putting  $z = \omega$ ,

$$(1+\omega^2+\omega^4)^8 = C_0 + C_1\omega^2 + C_2\omega^4 + \dots + C_{16}\omega^{32}$$

$$\Rightarrow C_0 + C_1\omega^2 + C_2\omega + C_3 + \dots + C_{16}\omega^2 = 0 \quad \dots(ii)$$

Putting  $z = \omega^2$ ,

$$(1+\omega+\omega^8)^8 = C_0 + C_1\omega^4 + C_2\omega^8 + \dots + C_{16}\omega^{64} \Rightarrow C_0 + C_1\omega + C_2\omega^2 + \dots + C_{16}\omega = 0 \quad \dots(iii)$$

Putting  $z = 1$ ,

$$3^8 = C_0 + C_1 + C_2 + \dots + C_{16} \quad \dots(iv)$$

$$\text{Adding (ii), (iii) and (iv), we have } 3(C_0 + C_3 + \dots + C_{15}) = 3^8 \Rightarrow C_0 + C_3 + \dots + C_{15} = 3^7$$

Similarly, first multiplying (i) by  $z$  then putting  $1, \omega, \omega^2$  and adding, we get:

Multiplying (i) by  $z^2$  and then putting  $1, \omega, \omega^2$  and adding, we get

$$C_2 + C_5 + C_8 + C_{11} + C_{14} = 3^7$$

12.(AB)

$$(a+b)x + (a-b)y - 2ab = 0 \text{ and } (a-b)x + (a+b)y - 2ab = 0$$

Equation of the angle bisectors are  $(a+b)x + (a-b)y - 2ab = 0 = \pm((a-b)x + (a+b)y - 2ab)$

$$2bx - 2by = 0 \text{ i.e., } x = y \text{ and } 2ax + 2ay - 4ab = 0 \text{ i.e., } x + y = 2b$$

$\therefore$  Equation of third side is given by

$$(i) \quad x - y = k \text{ satisfying the point } (b-a, a-b)$$

$$\therefore k = 2b - 2a \quad \therefore \text{ the line is } x - y = 2(b-a)$$

$$(ii) \quad x + y - 2b = k \text{ passing through the point } (b-a, a-b)$$

$$\therefore k = -2b \quad \therefore \text{ the line is } x + y = 0$$

$$1.(97) \quad N = \left( \sqrt[8]{5} + \sqrt[6]{2} \right)^{100}$$

$$T_{r+1} = {}^{100}C_r \left( \sqrt[8]{5} \right)^{100-r} \left( \sqrt[6]{2} \right)^r = {}^{100}C_r (5)^{\frac{100-r}{8}} (2)^{\frac{r}{6}}$$

For it to be rational  $r = 12, 36, 60, 84$

Hence, 4 rational terms & 97 irrational terms

$$2.(192) \quad 12 + 36 + 60 + 84 = 192$$

3.(0.50) & 4.(5)

Let  $A(x_1, y_1)$ ,  $B(x_2, y_2)$  and  $C(x_3, y_3)$  be the vertices of triangle ABC.

$\therefore$  Centroid

$$x_1 + x_2 + x_3 = 3 \times 3$$

$$x_1 + x_2 + x_3 = 9 \quad \dots(i)$$

$$\text{and } y_1 + y_2 + y_3 = 6 \quad \dots(ii)$$

Since, F is mid-point of AB,

$$\therefore x_1 + x_2 = 2 \quad \dots(iii)$$

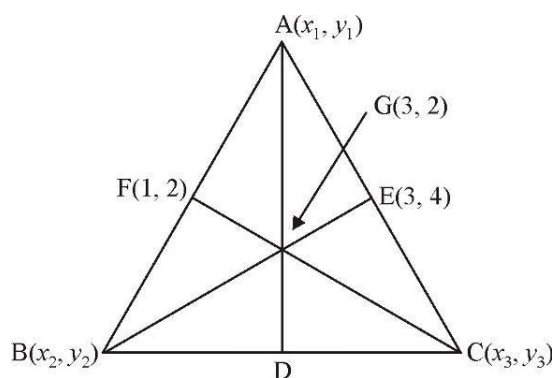
$$\text{and } y_1 + y_2 = 4 \quad \dots(iv)$$

Similarly, E is mid-point of AC

$$x_1 + x_3 = 6 \quad \dots(v)$$

$$y_1 + y_3 = 8 \quad \dots(vi)$$

On solving equation (i), (iii) and (ii), (iv), we get



$$x_3 = 7, y_3 = 2$$

On solving equations (i), (v) and (ii), (vi), we get

$$x_2 = 3, y_2 = -2$$

$\therefore$  Coordinates of B and C are (3, -2) and (7, 2) respectively

Equation of AB,

$$y - 2 = \frac{-4}{2}(x - 1)$$

$$y - 2 = -2(x - 1)$$

$$y - 2 = -2x + 2$$

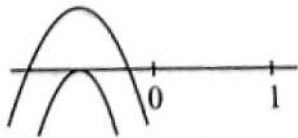
$$2x + y = 4$$

Coordinates of point D are  $\left(\frac{3+7}{2}, \frac{-2+2}{2}\right)$  i.e., (5, 0)

### Paragraph for Question 5-6

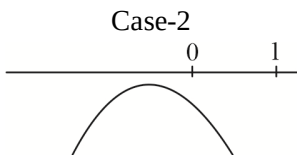
5.(5) **Case-1**  $c > 0$  not possible as  $g(x) = c(x^2 + 2x + 1) + (2x + 1) > 0, x \in (0, 1)$  and  $c > 0$

If  $c < 0$ , then for  $g(x) < 0 \forall x \in (0, 1)$ , vertex is negative.



$$D = 4(c+1) \geq 0, g(0) = (c+1) \leq 0 \quad \text{and} \quad \frac{-b}{2a} = \frac{-(c+1)}{c} \leq 0, g(1) = 3(c+1) + c < 0$$

Only possibility is  $c = -1$



$$D = 4(c+1) < 0, g(0) = (c+1) \leq 0 \quad \text{and} \quad \frac{-b}{2a} = \frac{-(c+1)}{c} \leq 0, g(1) = 3(c+1) + c < 0$$

$$c \in (-\infty, -1)$$

(6).(0.58)

**Case 1:** When both  $f(x)$  and  $g(x)$  are concave up,  $c > 1$

Possible graph to have  $f(x) \leq 0$  and  $g(x) \geq 0$  to have unique solution



Hence,  $D_f = 0 \Rightarrow 4c^2 - 4(c-1)(c+4) = 0$

Hence,  $c = \frac{4}{3}$

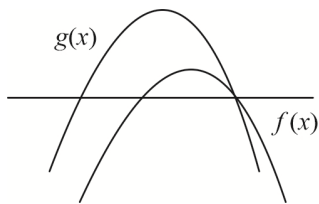
At  $c = \frac{4}{3}$ ,  $f(x) = (x+4)^2$  and at  $x = \frac{-c}{c-1} = -4$ ,  $g(-4) > 0$

$\therefore$  This case is possible. Hence,  $c = \frac{4}{3}$

**Case 2:**

Both  $f(x)$  and  $g(x)$  are concave down.,  $c < 0$

Possible graph to have  $f(x) \leq 0$  and  $g(x) \geq 0$  to have unique solution.



Here  $f(x)$  and  $g(x)$  have a common root.

In this case, the value of  $c$  is  $-\frac{3}{4}$ .

**Case 3:**

When one is concave up and another is concave down is rejected. Hence, all possible values of  $c$  are  $\frac{4}{3}$  and

$-\frac{3}{4}$ .

$\therefore \text{Sum} = \frac{4}{3} + \left(-\frac{3}{4}\right) = \frac{7}{12}$ .

## Solutions to JEE Advanced – 4 | JEE 2023 | Paper – 2 (Gen 1 to 6)

## PHYSICS

$$1.(ABC) \quad mgR \sin \theta = \frac{1}{2} mv^2$$

$$v = \sqrt{2gR \sin \theta}$$

$$N - mg \sin \theta = \frac{mv^2}{R}$$

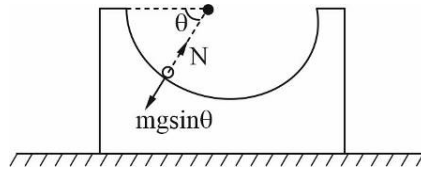
$$N = 3mg \sin \theta$$

$$f = N \cos \theta$$

$$= 3mg \sin \theta \cos \theta = \frac{3mg}{2} \sin 2\theta$$

$$a = \sqrt{\left(\frac{v^2}{R}\right)^2 + a_t^2} = \sqrt{4(g \sin \theta)^2 + g^2 \cos^2 \theta}$$

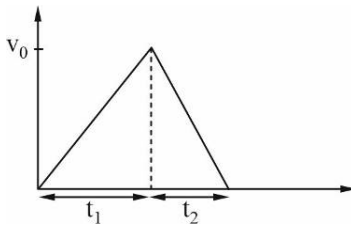
$$= g\sqrt{3\sin^2 \theta + 1}$$



$$2.(B) \quad F_{\text{base}} + F_{\text{side}} = mg + P_0 A_{\text{top}}$$

$$F_{\text{base II}} < F_{\text{base I}} \Rightarrow F_{\text{side II}} > F_{\text{side I}}$$

3.(ABC)



$$a_{\text{max}} = 20 \text{ m/s}^2$$

$$\text{Retardation (max)} = \frac{10m}{s^2}$$

$$\frac{v_0}{t_1} = 20 \quad \frac{v_0}{t_2} = 10$$

$$\frac{1}{2} v_0 (t_1 + t_2) = 30$$

$$\frac{v_0}{2} \left( \frac{v_0}{20} + \frac{v_0}{10} \right) = 30 \Rightarrow v_0 = 20 \text{ m/s} ; t_1 = 1\text{s} ; t_2 = 2\text{s}$$



4.(BC)

$$i = -KA \frac{d\theta}{dx}$$

$$i = \frac{-\alpha A}{\theta} \frac{d\theta}{dx}$$

$$i \int_0^\ell dx = -\alpha A \int_{2\theta_0}^{\theta_0} \frac{d\theta}{\theta}$$

$$i = \frac{\alpha A \ln(2)}{\ell}$$

$$ix = -\alpha A \ln\left(\frac{\theta}{2\theta_0}\right)$$

$$\ln\left(\frac{\theta}{2\theta_0}\right) = \frac{-ix}{\alpha A}$$

$$= -\ln(2) \frac{x}{\ell}$$

$$\frac{\theta}{2\theta_0} = 2^{-x/\ell}$$

$$\theta = 2\theta_0 2^{-x/\ell}$$

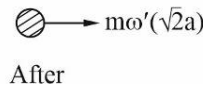
5.(AB)  $F_0 = 4m\omega_0^2 \frac{a}{\sqrt{2}} = 2\sqrt{2}m\omega_0^2 a$

Cons. of angular momentum about 'O'

$$(2ma^2 + m(\sqrt{2}a)^2)\omega_0 = (2ma^2 + 2m(\sqrt{2}a)^2)\omega'$$

$$4ma^2\omega_0 = 6ma^2\omega' \quad ; \quad \omega' = \frac{2\omega_0}{3}$$

$$F' = (5m)\omega'^2 \left( \frac{3\sqrt{2}a}{5} \right) = m \times \frac{4\omega_0^2}{9} \times 3\sqrt{2}a = \frac{4\sqrt{2}m\omega_0^2}{3} a$$



$$J = \frac{2\sqrt{2}m\omega_0 a}{3} \hat{i} - mv_0 \hat{j}$$

6.(AC) Shearing stress =  $\frac{F}{A}$

$$= \frac{\eta(2\pi R\ell)\omega_0 R}{h(2\pi R\ell)} = \frac{\eta\omega_0 R}{h} T_{\text{ext}} = \frac{\eta\omega_0 R}{h} (2\pi R\ell) \times R = \frac{2\pi\eta\omega_0 \ell R^3}{h} ; P_{\text{ext}} = T_{\text{ext}}\omega_0 = \frac{2\pi\eta\omega_0^2 \ell R^3}{h}$$

7.(C)

8.(A)

$$\Rightarrow |\vec{v}| = v_0 \quad \text{as } \vec{u} \perp \vec{v} \text{ at all times}$$

$$d\theta = \frac{|d\vec{v}|}{v_0} = \frac{u}{v_0} \frac{dm}{m} ; \quad \theta = \frac{u}{v_0} \ln(2)$$

9.(C) 10.(B)

Angular momentum is conserved about O

$$mv_0 \ell_0 = mv_{\perp} \left( \ell_0 + \frac{\ell_0}{5} \right)$$

$$v_{\perp} = \frac{5v_0}{6}$$

Conservation of energy

$$\frac{1}{2}mv_0^2 = \frac{1}{2}m \left( \frac{25v_0^2}{36} \right) + \frac{1}{2}k \frac{\ell_0^2}{25}$$

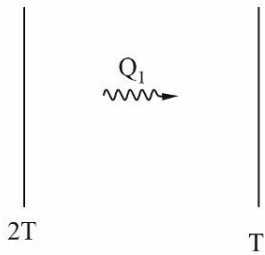
$$\frac{k\ell_0^2}{25} = mv_0^2 \left( 1 - \frac{25}{36} \right) = \frac{11mv_0^2}{36}$$

$$k = \frac{11mv_0^2}{36} \times \frac{25}{\ell_0^2} = 275 \frac{N}{m}$$

11.(D)

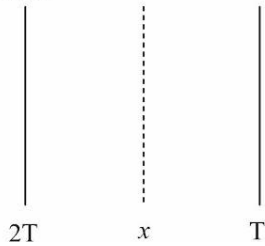
12.(C)

Before



$$Q_1 = \sigma A(2T)^4 - \sigma A(T)^4 = 15\sigma AT^4$$

After

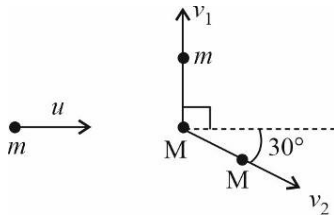


$$\frac{1}{2}(\sigma A(2T)^4 + \sigma AT^4) = \frac{2\sigma Ax^4}{2} \Rightarrow x^4 = \frac{17}{2}T^4 \Rightarrow x = \left( \frac{17}{2} \right)^{\frac{1}{4}} T$$

$$Q_2 = \sigma A(2T)^4 - \frac{\sigma Ax^4}{2} - \frac{\sigma AT^4}{2}$$

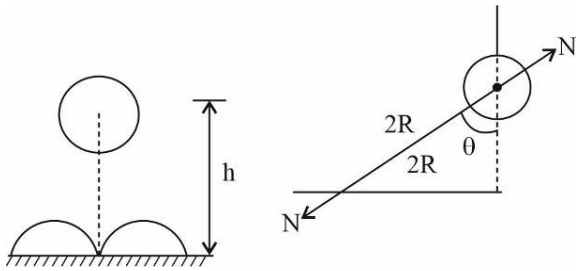
$$= \sigma AT^4 \left( 16 - \frac{17}{4} - \frac{1}{2} \right) = \sigma AT^4 \left( \frac{64 - 17 - 2}{4} \right) = \sigma AT^4 \left( \frac{45}{4} \right) \Rightarrow \frac{Q_1}{Q_2} = \frac{15}{45} \times 4 = \frac{4}{3}$$

1.(5)



By conservation of momentum  $mu = Mv_2 \frac{\sqrt{3}}{2}$  ;  $\frac{Mv_2}{2} = mv_1$

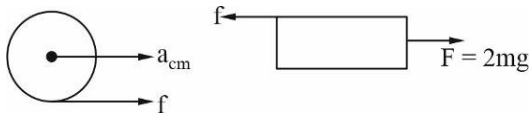
2.(9)  $\sin \theta = \frac{1}{\sqrt{3}}$



$$\therefore \cos \theta = \sqrt{\frac{2}{3}} \quad \therefore \int 3N \cos \theta dt = mv_0 \quad ; \quad \int N \sin \theta dt = \frac{m}{2} v$$

$$\therefore 3 \cot \theta = \frac{2v_0}{v} \quad \therefore v = \frac{2}{3\sqrt{2}} v_0 = \frac{\sqrt{2}}{3} v_0 \quad \therefore p = \frac{\frac{1}{2}mv_0^2}{\frac{1}{2} \cdot \frac{m}{2} \cdot \frac{2v_0^2}{9}} = 9$$

3.(1.11) The free body diagrams of sphere A and Block B are as shown



Applying Newton's law to block and sphere

$$F - f = ma \quad \dots (i)$$

$$f = ma_{cm} \quad \dots (ii)$$

$$fr = \frac{2}{5}mr^2\alpha \quad \dots (iii)$$

$$\text{For no slipping, } a = a_{cm} + r\alpha \quad \dots (iv)$$

$$\therefore \alpha = \frac{10g}{9r}$$

4.(14.42)

Collision will be perpendicular to incline surface hence

$$(1) \quad x = u_x t - \frac{1}{2} g \sin \theta \cdot t^2$$

$$(2) \quad o = u_y t - \frac{1}{2} g \cos \theta \cdot t^2$$

$$(3) \quad u_x = g \sin \theta \cdot t$$

$$u = \sqrt{u_x^2 + u_y^2}$$

$$5.(3) \quad -1 = 3t_0 - \frac{1}{2}(8)t_0^2 \Rightarrow t_0 = 1s$$

$$\text{Disp.} = 8(1) - \frac{1}{2}(10)(1)^2 = 3m$$

6.(17.32)

But  $\vec{V}_{R/G(x)} = 0$ , since rain fall vertically down.

$$\Rightarrow \quad \vec{V}_{C/G(x)} = -V_{R/C(x)} = -(V \cos 30^\circ)$$

$$\Rightarrow \quad |\vec{V}_{C/G(x)}| = V \cos 30^\circ \quad \dots (i)$$

Now,  $\vec{V}_{R/G(y)} = \vec{V}_{R/C(y)}$ 

$$\Rightarrow \quad -10 = -V \sin 30^\circ + 0$$

[Since :  $V_{R/G(y)} = -10m/s$ ;  $V_{C/G(y)} = 0$ ]

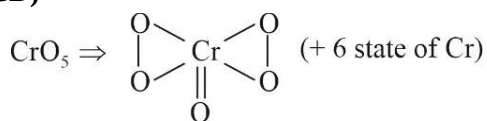
$$\Rightarrow \quad V \sin 30^\circ = 10 \Rightarrow V = 20 m/s$$

Substituting  $V = 20m/s$  in equation (i)

$$\vec{V}_{C/G(x)} = V \cos 30^\circ = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3} m/s$$

$$\therefore \quad \vec{V}_{C/G} = 10\sqrt{3}i m/s$$

1.(ABCD)

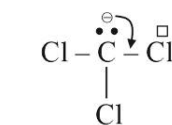


$$\text{Cr}(\text{O}_2)_4^{3-}, x - 8 = -3, x = +5$$

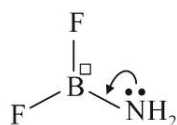
$$\text{CrO}_4^{3-}, x - 8 = -3, x = +5$$

$$[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3, x - 3 = 0, x = +3$$

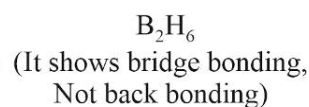
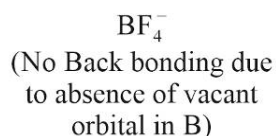
2.(CD)



(Show Back bonding)



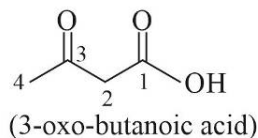
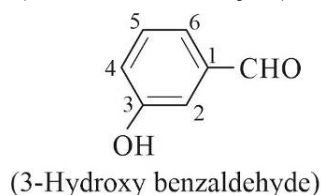
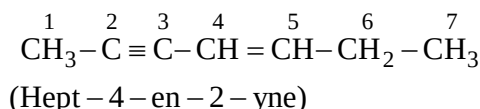
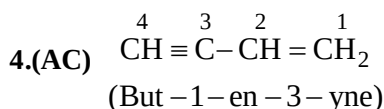
(Back bonding)



3.(ABC)

Order given in option A, B and C are correct

# Li have greater E. A than Be (Stable fully filled electronic configuration).



5.(BCD)

For the gas, Vander Waal's constant "a" = 0 but "b" ≠ 0. Thus statements of option B, C and D are valid for the gas.

6.(ABCD)

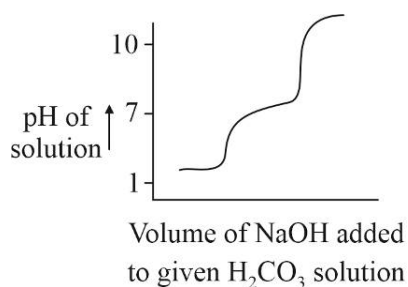
$$\text{Given that } K_C > 1, \Rightarrow \Delta_r G^\circ < 0$$

As yield of the product increases with temperature thus  $\Delta_r H^\circ > 0$ . Now as  $\Delta_r G^\circ = \Delta_r H^\circ - T\Delta_r S^\circ$ ; for  $\Delta_r G^\circ$  to be negative when  $\Delta_r H^\circ > 0$ ,  $\Delta_r S^\circ$  must be positive.

7.(C) Reaction (C) only is a disproportionation redox reaction

8.(C) Given in option A, B and D are comproportionation redox reaction.

9.(C)



This pH curve will have 2 nearly vertical portions as  $\text{H}_2\text{CO}_3$  is a diprotic acid

10.(D) pH of the solution at half neutralization point =  $\text{pK}_a$ , by the given figure  $\text{pK}_a = 6$

Thus  $K_a = 10^{-6}$

11.(C)  $\Delta S = \Delta S_{\text{univ}} + \Delta S_{\text{sys}} + \Delta S_{\text{surr.}} = (+)\text{ve}$  only in option C

$$12.(C) \Delta S_{\text{univ}} = -\frac{\Delta G}{T} = -\left[\frac{\Delta H_{\text{sys}} - T\Delta S_{\text{sys}}}{T}\right] \left[\frac{1}{T}\right]$$

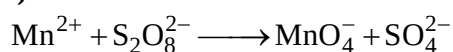
$$= -\frac{\Delta H_{\text{sys}}}{T} + \Delta S_{\text{sys}} = -\frac{20 \times 1000}{400} + 100 = -50 + 100 = +50$$

1.(8) Except  $\text{NH}_3$ , each of the species are formed by using certain excited state of the central atom for hybridization.

2.(83)

$$\begin{aligned} E_{(\text{photon})} &= \phi + K.E \\ \phi &= 13.6 \text{ eV} \\ K.E &= 1.4 \text{ eV} \\ E_{(\text{photon})} &= 15 \text{ eV} \end{aligned} \quad \left| \begin{aligned} 15 \times 1.6 \times 10^{-19} \text{ J} &= \frac{hc}{\lambda} \\ \lambda &= \frac{6.67 \times 3 \times 10^{-26}}{24 \times 10^{-19}} \\ &\approx \frac{20 \times 10^{-26}}{24 \times 10^{-19}} = 83.375 \text{ nm} \end{aligned} \right.$$

3.(0.40)



$n_f$  of  $\text{Mn}^{2+} = 5$ ;  $n_f$  of  $\text{S}_2\text{O}_8^{2-} = 2$

By law of chemical equivalence.

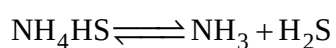
$5 \times \text{moles of } \text{Mn}^{2+} = 2 \times \text{moles of } \text{S}_2\text{O}_8^{2-}$

Moles of  $\text{Mn}^{2+} = 0.4$  mole (per mole of  $\text{S}_2\text{O}_8^{2-}$ )

$$4.(0.42) K_{\text{sp}} \times K_f(\text{complex}) = \frac{S^2}{[\text{NH}_3]^2} = \frac{x^2}{(5-2x)^2} = 10^{-2}$$

$$\Rightarrow \frac{x}{(5-2x)} = 0.1 \text{ or } x = 0.5 - 0.2x \Rightarrow 1.2x = 0.5 \text{ or } x = 0.41667$$

5.(5.25)

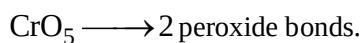
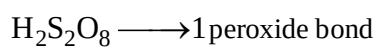
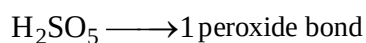
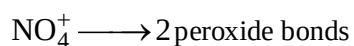
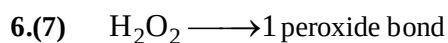


$$\begin{array}{ccc} 2 & 0 & \\ (2+P) & P & \end{array}$$

$$P_T = 5 = 2 + 2P$$

$$\Rightarrow 2P = 3 \text{ (or) } P = 1.5 \text{ atm.}$$

$$\begin{aligned} K_p &= P_{\text{NH}_3} P_{\text{H}_2\text{S}} \\ &= (3.5)(1.5) = 5.25 \end{aligned}$$



## MATHEMATICS

1.(ABCD)

$P(2, -1)$  goes 2 units along  $x + y = 1$  upto A and 5 units along  $x - 2y = 4$  upto B.

Slope of PA  $= -1 = \tan 135^\circ$

$$\text{Slope of PB} = \frac{1}{2} = \tan \theta \Rightarrow \sin \theta = \frac{1}{\sqrt{5}}, \cos \theta = \frac{2}{\sqrt{5}}$$

The coordinates of B

$$\text{i.e., } (x_1 + r \cos \theta, y_1 + r \sin \theta) \text{ and } (2\sqrt{5} + 2, \sqrt{5} - 1)$$

$$\text{The coordinates of A i.e., } (x_1 + r \cos 135^\circ, y_1 + r \sin 135^\circ) \text{ and } (2 - \sqrt{2}, \sqrt{2} - 1)$$

## 2.(ABCD)

$$(A) \quad \log_2 3 - \log_5 17 = \frac{\log_2 3 \log_2 5 - \log_2 17}{\log_2 5}$$

$$\text{But } \log_2 17 > 4$$

$$\therefore 2^5 > 5^2 \Rightarrow 2^{5/2} > 5 \Rightarrow \frac{5}{2} > \log_2 5$$

$$\log_2 5 < \frac{5}{2}; 3^5 < 2^8; 5 \log_2 3 < 8; \log_2^3 < \frac{8}{5}$$

$$\therefore \log \frac{5}{2}, \log \frac{3}{2} < 4 \quad \therefore \log_2^3 \cdot \log_2^5 - \log_2^{17} < 0 \Rightarrow \log_2^3 < \log_5^{17}$$

$$(B) \quad p = \log_{12} 18 = \frac{\log_2 18}{\log_2 12} = \frac{1 + 2 \log_2 3}{2 + \log_2 3} \text{ and}$$

$$q = \log_{24} 54 = \frac{\log_2 54}{\log_2 24} = \frac{1 + 3 \log_2 3}{3 + \log_2 3}$$

$$\text{Putting } x = \log_2 3, \quad pq + 5(p - q)$$

$$= \frac{1+2x}{2+x} \cdot \frac{1+3x}{3+x} + 5 \left\{ \frac{1+2x}{2+x} - \frac{1+3x}{3+x} \right\} = \frac{6x^2 + 5x + 1 + 5(-x^2 + 1)}{(x+2)(x+3)} = \frac{x^2 + 5x + 6}{(x+2)(x+3)} = 1$$

$$(C) \quad \text{Let } x = \log_2 5 > 2$$

$$x^2 - x - 2 = (x-2)(x+1) > 0 \because x > 2$$

$$\therefore x^2 - x - 2 > 0 \Rightarrow (\log_2^5)^2 > \log_2^5 + 2 = \log_2^{20}$$

$$(D) \quad \log_{3/4} \left[ \frac{1}{3} \log_2 (x^2 + 7) \right] + \log_{1/2} \left[ \frac{1}{2} \log_2 (x^2 + 7) \right] = -2$$

$$\text{Let } \log_2 (x^2 + 7) = t \quad \therefore x^2 + 7 = 2^t$$

$$\log_{3/4} \left( \frac{1}{3} t \right) + \log_{1/2} \left( \frac{1}{2} t \right) = -2$$

$$\frac{\log t - \log 3}{\log 3 - \log 4} + \log_{1/2} \frac{1}{2} + \log_{1/2} t = -2$$



$$\frac{\log t}{\log 3 - 2 \log 2} - \frac{\log 3}{\log 3 - 2 \log 2} + 1 - \frac{\log t}{\log 2} = -2$$

$$\therefore \log t \frac{[\log 2 - \log 3 + 2 \log 2]}{(\log 3 - 2 \log 2) \log 2} = -3 + \frac{\log 3}{\log 3 - 2 \log 2}$$

$$\log t \frac{[3 \log 2 - \log 3]}{(\log 3 - 2 \log 2) \log 2} = \frac{6 \log 2 - 2 \log 3}{\log 3 - 2 \log 2}$$

$$\therefore \log t = 2 \log 2 = \log 2^2 = \log 4$$

$$\therefore t = 4 \therefore x^2 + 7 = 2^t = 2^4 = 16 \text{ or } x^2 = 9 \therefore x = \pm 3$$

Important (no need to solve either there will be no solution or 2 solutions)

3.(BD) Suppose that the first term of the sequence  $\Delta A$  is  $d$ .

Then

$$\Delta A = \{d, d+1, d+2, \dots\} \text{ with the } n^{\text{th}} \text{ term given by } d + (n-1).$$

Hence,  $A = \{a_1, a_1 + d, a_1 + d + (d+1), a_1 + d + (d+1) + (d+2), \dots\}$  with the  $n^{\text{th}}$  term given by

$$a_n = a_1 + (n-1)d + \frac{1}{2}(n-1)(n-2).$$

This shows that  $a_n$  is a quadratic polynomial in  $n$  with leading coefficient  $1/2$ .

Since  $a_{19} = a_{92} = 0$ , we must have

$$a_n = \frac{1}{2}(n-19)(n-92),$$

$$\text{So } a_1 = (1-19)(1-92)/2 = 819.$$

$$a_{42} = -575$$

$$s_5 = \frac{1}{2} \sum_{n=1}^5 (n^2 - 111n + 19 \times 92)$$

$$= \frac{1}{2} \left[ \frac{5 \times 6 \times 11}{6} - \frac{111 \times 5 \times 6}{2} + 19 \times 92 \times 5 \right] = \frac{1}{2} [55 - 111 \times 15 + 19 \times 92 \times 5]$$

$$= \frac{5}{2} [11 - 333 + 19 \times 92] = 5 \times 713 = 5 \times 23 \times 31 = 3565$$

$$4.(D) \left( \frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - x^{1/2}} \right)^{10} = \left( x^{1/3} + 1 - 1 - \frac{1}{\sqrt{x}} \right)^{10}$$

$$T_{r+1} = {}^{10}C_r \left(x^{1/3}\right)^{10-r} \left(-\frac{1}{\sqrt{x}}\right)^r$$

$$\text{For independent term } \frac{10-r}{3} - \frac{r}{2} = 0 \Rightarrow r = 4$$

$$\text{Coefficient of the term independent of } x = {}^{10}C_4$$

$$5.(ABD) \quad a_1 + a_4 + a_7 + \dots + a_{16} = 147$$

$$3(a_1 + a_{16}) = 147$$

$$a_1 + a_{16} = 49$$

$$\text{Also, } a_1 + a_4 + a_7 + a_{10} + \dots + a_{16} = a_1 + a_1 + 3d + a_1 + 6d + \dots + a_1 + 15d = 6a_1 + 45d = 147 = 2a_1 + 15d = 49$$

$$a_1 + a_6 + a_{11} + a_{16} = a_1 + a_1 + 5d + a_1 + 10d + a_1 + 15d = 4a_1 + 30d = 2(2a_1 + 15d) = 2(49) = 98$$

$$a_1 + a_{16} = a_1 + a_1 + 15d = 2a_1 + 15d = 49$$

Using  $AM \geq GM$ .

$$\frac{a_1 + a_2 + \dots + a_{16}}{16} \geq (a_1 a_2 a_3 \dots a_{16})^{1/16}$$

$$\frac{8(a_1 + a_{16})}{16} \geq (a_1 a_2 a_3 \dots a_{16})^{1/16}$$

$$\left(\frac{8(49)}{16}\right)^{16} \geq a_1 a_2 a_3 \dots a_{16}$$

$$\left(\frac{49}{2}\right)^{16} \geq a_1 a_2 a_3 \dots a_{16}$$

$$a_1 + a_4 + a_{11} + a_{16} = a_1 + a_1 + 3d + a_1 + 10d + a_1 + 15d = 4a_1 + 28d$$

$$6.(AD)$$

$$2P_6 - 3P_4 = 2(\cos^6 \theta + \sin^6 \theta) - 3(\cos^4 \theta + \sin^4 \theta)$$

$$= 2\left[(\cos^2 \theta + \sin^2 \theta)^3 - 3\sin^2 \theta \cos^2 \theta (\cos^2 \theta + \sin^2 \theta)\right]$$

$$- 3\left[(\cos^2 \theta + \sin^2 \theta)^2 - 2\sin^2 \theta \cos^2 \theta\right]$$

$$= 2(1 - 3\sin^2 \theta \cos^2 \theta) - 3(1 - 2\sin^2 \theta \cos^2 \theta) = -1$$

Now, for  $n \geq 4$  we have.

$$P_n - P_{n-2} = (\cos^n \theta + \sin^n \theta) - (\cos^{n-2} \theta + \sin^{n-2} \theta)$$

$$= \cos^{n-2} \theta (\cos^2 \theta - 1) + \sin^{n-2} \theta (\sin^2 \theta - 1)$$

$$= -\sin^2 \theta \cos^{n-2} \theta - \cos^2 \theta \sin^{n-2} \theta = -\sin^2 \theta \cos^2 \theta (\cos^{n-4} \theta + \sin^{n-4} \theta)$$

$$P_n - P_{n-2} = -\sin^2 \theta \cos^2 \theta P_{n-4} \quad \dots(i)$$

$$\begin{aligned}
 6P_{10} - 15P_8 + 10P_6 &= 6(P_{10} - P_8) - 9(P_8 - P_6) + (P_6 - P_4) + (P_4 - P_2) + P_2 \\
 &= -\sin^2 \theta \cos^2 \theta \{6P_6 - 9P_4 + P_2 + P_0\} + 1 \\
 &= -3\sin^2 \theta \cos^2 \theta \{2P_6 - 3P_4\} - \sin^2 \theta \cos^2 \theta (1+2) + 1 \\
 &= 3\sin^2 \theta \cos^2 \theta - 3\sin^2 \theta \cos^2 \theta + 1 = 1
 \end{aligned}$$

- 7.(B) We have,  $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$   
 Put  $x = 1$   
 $3^n = a_0 + a_1 + a_2 + a_3 + \dots + a_{2n}$   
 $3^n = (a_0 + a_{2n} + a_1 + a_{2n-1} + \dots + a_{n-1} + a_{n+1}) + a_n$   
 $3^n = 2(a_0 + a_1 + a_2 + \dots + a_{n-1}) + a_n \quad [\because a_r = a_{2n-r}]$   
 $\therefore$  Hence, option (B) is correct.

- 8.(A) Put  $x = \frac{-1}{x}$  in place of  $x$

$$\left(1 - \frac{1}{x} + \frac{1}{x^2}\right)^n = a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} + \dots + \frac{a_{2n}}{x^{2n}} \quad \dots(i)$$

$$\Rightarrow (x^2 - x + 1)^n = a_0x^{2n} - a_1x^{2n-1} + a_2x^{2n-2} + \dots + a_{2n} \quad \dots(ii)$$

On multiplying equation (i) and (ii) we get

$$(1+x^2+x^4)^n = (a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n})(a_0x^{2n} - a_1x^{2n-1} + \dots + a_{2n}) \quad \dots(iii)$$

On putting  $x^2$  in place of  $x$ , we get

$$(1+x^2+x^4)^n = a_0 + a_1x^2 + a_2x^4 + \dots + a_nx^{2n} + \dots + a_{2n}x^{4n}$$

Coefficient of  $x^{2n}$  in  $(1+x^2+x^4)^n$  is  $a_n$ .

On equating the coefficient of  $x^{2n}$ , we get  $a_0^2 - a_1^2 + a_2^2 - a_1^2 - \dots + a_{2n}^2 = a_n$  Hence,  $k = 1$

- 9.(A) Let  $f(x) = (1+x)^{2000} = \sum_{k=0}^{2000} \binom{2000}{k} x^k$

Let  $\omega = (-1 + \sqrt{3}i)/2$ . Then  $\omega^3 = 1$  and  $\omega^2 + \omega + 1 = 0$ . Hence

$$\begin{aligned}
 &3 \left( \binom{2000}{2} + \binom{2000}{5} + \dots + \binom{2000}{2000} \right) \\
 &= f(1) + \omega f(\omega) + \omega^2 f(\omega^2) = 2^{2000} + \omega(1+\omega)^{2000} + \omega^2(1+\omega^2)^{2000} \\
 &= 2^{2000} + \omega(-\omega^2)^{2000} + \omega^2(-\omega)^{2000} = 2^{2000} + \omega^2 + \omega = 2^{2000} - 1
 \end{aligned}$$

Thus, the desired value is  $\frac{2^{2000} - 1}{3}$

- 10.(A)  $\sum_{k=0}^7 \left( \frac{{}^7C_k}{{}^{14}C_k} \sum_{r=k}^{14} r {}^rC_k {}^{14}C_r \right)$

$$= \sum_{k=0}^7 \left( \frac{{}^7C_k}{14} \times \frac{k \cdot 14 - k}{14} \sum_{r=k}^{14} \frac{r}{k \cdot r - k} \cdot \frac{14}{r \cdot 14 - r} \right)$$

$$= \sum_{k=0}^7 \left( {}^7C_k \sum_{r=k}^{14} {}^{14-k}C_{r-k} \right) = \sum_{k=0}^7 {}^7C_k \cdot 2^{14-k} = 2^{14} \sum_{k=0}^7 {}^7C_k \left( \frac{1}{2} \right)^k = 2^{14} \cdot \left( 1 + \frac{1}{2} \right)^7 = 6^7 > 7^6$$

11.(D) Given lines  $3x + 4y = 5$  ... (i)  
 $4x - 3y = 15$  ... (ii)

$\therefore$  Lines are perpendicular

$\triangle ABC$  is right angled isosceles triangle with  $AB = AC$

$$\angle ABC = \angle ACB = 45^\circ$$

BC is equally inclined with AB and AC

Slope of line (i)  $m_1 = -3/4$

Slope of line (ii)  $m_2 = 4/3$

$$\left| \frac{m - 4/3}{1 + \frac{4}{3}m} \right| = \tan 45^\circ = 1$$

$$\left( m - \frac{4}{3} \right)^2 = \left( 1 + \frac{4}{3}m \right)^2$$

$$m^2 + \frac{16}{9} - \frac{8m}{3} = 1 + \frac{16m^2}{9} + \frac{8m}{3}$$

$$\frac{7m^2}{9} + \frac{16m}{3} - \frac{7}{9} = 0$$

$$7m^2 + 48m - 7 = 0$$

$$7m^2 + 49m - m - 7 = 0$$

$$7m(m+7) - 1(m+7) = 0$$

$$m = \frac{1}{7}, m = -7$$

$\therefore$  Required lines are

$$y - 2 = \frac{1}{7}(x - 1)$$

$$7y - 14 = x - 1$$

$$l_1 : x - 7y + 13 = 0$$

and  $y - 2 = -7(x - 1)$

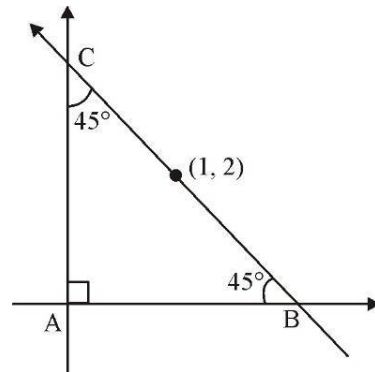
$$y - 2 = -7x + 7$$

$$l_2 : 7x + y - 9 = 0$$

Also,  $l_1 = x + ay + b = 0$  and  $l_2 : 7x + cy + d = 0$

On comparing  $a = -7, b = 13, c = 1, d = -9$

$$b + d = 13 - 9 = 4$$



12.(A) Co-ordinates of point P are  $(2b + d - 1)$  i.e.,  $P(2, 3)$

$\therefore$  Angle with positive x-axis  $= (90 + 60)^\circ = 150^\circ$

Let required co-ordinates be  $(x, y)$

$$\frac{x-2}{\cos 150^\circ} = \frac{y-3}{\sin 150^\circ} = r$$

$$x = 2 + r \cos 150^\circ$$

$$y = 3 + r \sin 150^\circ$$

$$r = 4 \quad \therefore x = 2 - 4 \frac{\sqrt{3}}{2} = 2 - 2\sqrt{3} ; \quad y = 3 + 4 \times \frac{1}{2} = 5$$

1.(816)

Let  $f(x)$  denote the given expression.

$$\text{Hence } f(x) = f(y-1) = \frac{1-(y-1)^{18}}{1+(y-1)} = \frac{1-(y-1)^{18}}{y}$$

Therefore  $a_2$  is equal to the coefficient of  $y^3$  in the expansion of  $1-(y-1)^{18}$ ,  $a_2 = \binom{18}{3} = 816$

2.(2) It is clear that  $x \geq -2$ . We consider the following cases

(i)  $-2 \leq x \leq 2$ . Setting  $x = 2 \cos a$ ,  $0 \leq a \leq \pi$ , the equation becomes

$$8 \cos^3 a - 6 \cos a = \sqrt{2(\cos a + 1)} \quad \text{Or} \quad 2 \cos 3a = \sqrt{4 \cos^2 \frac{a}{2}}$$

From which it follows that  $\cos 3a = \cos \frac{a}{2}$

$$\text{Then } 3a - \frac{a}{2} = 2m\pi, m \in \mathbb{Z}, \text{ or } 3a + \frac{a}{2} = 2n\pi, n \in \mathbb{Z}$$

Since  $0 \leq a \leq \pi$ , the solution in this case is  $x = 2 \cos 0 = 2$ ,  $x = 2 \cos \frac{4\pi}{5}$ , and  $x = 2 \cos \frac{4\pi}{7}$

(ii)  $x > 2$ . Then  $x^3 - 4x = x(x^2 - 4) > 0$  and  $x^2 - x - 2 = (x-2)(x+1) > 0$

$$x^2 - x - 2 = (x)^2 - (\sqrt{x+2})^2 > 0 \quad x > \sqrt{x+2}$$

It follows that  $x^3 - 3x > x > \sqrt{x+2}$

Hence there are no solution in this case

Therefore,  $x = 2$ ,  $x = 2 \cos 4\pi/5$ , and  $x = 2 \cos 4\pi/7$

3. (105)

$$8x^3 - 4x^2 - 4x + 1 = 0 \quad \text{--- (1)}$$

$$x \rightarrow \frac{1}{x} \Rightarrow x^3 - 4x^2 - 4x + 8 = 0 \quad \text{-- (2)}$$

$$\sec \frac{\pi}{7} + \sec \frac{3\pi}{7} + \sec \frac{5\pi}{7} = 4$$

Replacing  $x^2 \rightarrow x+1$  in (2)

$$\Rightarrow x(x^2 - 4) = 4(x^2 - 2) \Rightarrow \sqrt{x+1}(x+1-4) = 4(x+1-2)$$

$$\Rightarrow x^3 - 21x^2 + 35x - 7 = 0 \quad \text{--- (3)}$$

$$\tan^2 \frac{\pi}{7} + \tan^2 \frac{3\pi}{7} + \tan^2 \frac{5\pi}{7} = \frac{21}{1} = 21$$

$$x \rightarrow \frac{1}{x} \text{ in (3)} \Rightarrow 7x^3 - 35x^2 + 21x - 1 = 0 ; \cot^2 \frac{\pi}{7} + \cot^2 \frac{3\pi}{7} + \cot^2 \frac{5\pi}{7} = \frac{35}{7} = 5$$

$$\left( \sum_{r=1}^3 \tan^2 \left( \frac{2r-1}{7} \pi \right) \right) \left( \sum_{r=1}^3 \cot^2 \left( \frac{2r-1}{7} \pi \right) \right) = 21 \times 5 = 105$$

4.(6) Total number of digits = 9

Now, we can select 2 places for the digit 1 and 2 in  ${}^9C_2$  ways, from remaining 7 places select any two places for 3 and 4 in  ${}^7C_2$  ways and from the remaining 5 places, select any 3 for 5, 6 and 7 in  ${}^5C_3$  ways. Arrange 6 & 7 in 2 ways.

Now we are left with digits 8 & 9 and 2 places can be filled in 2! Ways.

$$\therefore \text{Total number of ways} = {}^9C_2 \cdot {}^7C_2 \cdot {}^5C_3 \cdot 2 \times 2 = 6 \times 7!$$

$$\therefore k = 6$$

5.(6) Let  $\tan x = a, \tan y = b, \tan z = c$

$$\text{Given system of equation is equal to } a + b + c = 6 - \frac{1}{a} - \frac{1}{b} - \frac{1}{c}$$

$$a^2 + b^2 + c^2 = 6 - \frac{1}{a^2} - \frac{1}{b^2} - \frac{1}{c^2}; \quad a^3 + b^3 + c^3 = 6 - \frac{1}{a^3} - \frac{1}{b^3} - \frac{1}{c^3}$$

$$\text{From second equation we complete squares to get } \left( a - \frac{1}{a} \right)^2 + \left( b - \frac{1}{b} \right)^2 + \left( c - \frac{1}{c} \right)^2 = 0 \Rightarrow a = b = c = \pm 1$$

Now rearranging 3<sup>rd</sup> equation and adding 3 times first equation, we get

$$a^3 + b^3 + c^3 + \frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} + 3 \left( a + \frac{1}{a} + b + \frac{1}{b} + c + \frac{1}{c} \right) = 6 + 18$$

$$\left( a + \frac{1}{a} \right)^3 + \left( b + \frac{1}{b} \right)^3 + \left( c + \frac{1}{c} \right)^3 = 24$$

Therefore  $a = b = c = 1$

$$\text{Hence, } \left[ \frac{\tan(x)}{\tan(y)} + \frac{\tan(y)}{\tan(z)} + \frac{\tan(z)}{\tan(x)} + 3 \tan(x) \tan(y) \tan(z) \right] = 6$$

$$6.(1) (5 + 2\sqrt{6})^n = p + f$$

$$(5 - 2\sqrt{6})^n = f' \Rightarrow 0 < f + f' < 2$$

$$p + f + f' = 2 \times (\text{Integer})$$

$$\text{So } f + f' = \text{integer} = 1 \quad \because p \in N$$

$$(f - 1)(p + f) = -f'(p + f) = -(+1)^n = -1$$